Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 11

Roll No : 310

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Progrmas based on Python Data Structure + UDF + File Handling

Practical 1: Count Lines, Words and Characters

Problem Definition:

Write a Python program to read text from a file and count the total number of lines, words, and characters. The result should be written to an output file.

In [8]:

```
lines=0;words=0;chars=0
with open("test.txt","r") as fp :
    for line in fp:
        lines +=1
        words += len(line.split())
        chars += len(line)
with open("output.txt","w") as fp :
    fp.write(f"{lines}\n")
    fp.write(f"{words}\n")
    fp.write(f"{chars}\n")
print("Count successfully written to output file.")
```

```
Count successfully written to output file.
```

Practical 2: Search a Word in a File

Problem Definition:

Write a Python program to search a given word in a text file and count its occurrences. Store the result in an output file.

In [11]:
```
search_word = input("Enter the word")
count=0
with open("test.txt","r") as fp :
    for line in fp:
        words = line.split()
        for word in words:
            if word == serach_word:
                count+=1
with open("output.txt","w") as fp:
    fp.write(f"{search_word}\n")
    fp.write(f"{count}\n")
print("Search result written to output file.")
```

```
Search result written to output file.
```

Practical 3: Student Marks and Average

Problem Definition:

Write a Python program to read roll number, name, and marks of students from a file. Store marks in a list, calculate the average using a user-defined function, and write student details along with average marks to an output file.

In [14]:
```
def cal_average(marks):
    return sum(marks)/len(marks)
marks_list = []
students = []
with open("student.txt","r") as fp:
    for line in fp:
        roll,name,marks = line.split()
        marks = int(marks)
        marks_list.append(marks)
        students.append((roll,name,marks))
avg = cal_average(marks_list)
with open("result.txt","w") as fp:
    fp.write("RollNo  Name  Marks\n")
    for s in students:
        fp.write(f"{s[0]}      {s[1]}     {s[2]}\n")
    fp.write(f"{avg}")
print("Student details and average written to output file.")
```

```
Student details and average written to output file.
```

Practical 4: Maximum and Minimum using Tuple

Problem Definition:

Write a Python program to read numbers from a file, store them in a tuple, and find the maximum and minimum values. Write the result to an output file.

In [15]:
```
with open("nums.txt","r") as fp:
    numbers = fp.read().split()
nums = tuple(int(num) for num in numbers)
max1 = max(nums)
min1 = min(nums)
with open("result.txt","w") as fp:
    fp.write(f"{nums}\n")
    fp.write(f"{max1}\n")
```

```python
    fp.write(f"{min1}\n")
print("Maximum and Minimum written to output file.")
```

```
Maximum and Minimum written to output file.
```

Practical 5: Remove Duplicate Words using Set

Problem Definition:

Write a Python program to read words from a file and remove duplicate words using a set. Store the unique words in an output file.

In [16]:
```python
with open("demo.txt","r") as fp:
    words = fp.read().split()
unique_words = set(words)
with open("test.txt","w") as fp:
    for word in unique_words:
        fp.write(word+" ")
print("Duplicate words removed and saved to output file.")
```

```
Duplicate words removed and saved to output file.
```

Practical 6: Word Frequency using Dictionary

Problem Definition:

Write a Python program to count the frequency of each word in a text file using a dictionary and write the result to an output file.

In [4]:
```python
file = open("input.txt", "r")
text = file.read()

text = text.lower()

words = text.split()

word_count = {}

for word in words:
    if word in word_count:
        word_count[word] = word_count[word] + 1
    else:
        word_count[word] = 1

file = open("output.txt", "w")

for word in word_count:
    file.write(word + " : " + str(word_count[word]) + "\n")

file.close()
print("Done! Check output.txt")
```

```
Done! Check output.txt
```

Practical 7: Student Result and Grade Calculation

Problem Definition:

Write a Python program to read roll number, name, and marks of students from a file.

Use a user-defined function to determine grade (A/B/C/Fail) and write the student result to an output file.

Grade Criteria:

- Marks ≥ 75 : Grade A
- Marks ≥ 60 : Grade B
- Marks ≥ 40 : Grade C
- Marks < 40 : Fail

In [6]:

```
def get_grade(marks):
    if marks >= 75:
        return "A"
    elif marks >= 60:
        return "B"
    elif marks >= 40:
        return "C"
    else:
        return "Fail"

file = open("students.txt", "r")
out = open("result.txt", "w")

for line in file:
    data = line.split()
    roll = data[0]
    name = data[1]
    marks = int(data[2])

    grade = get_grade(marks)

    out.write(roll + " " + name + " " + str(marks) + " " + grade + "\n")

file.close()
out.close()

print("Result written to result.txt")
```

```
Result written to result.txt
```

In [ ]: